HONG CHING YUAN

Engineering Entrepreneurship Portfolio & FYP Showcase

UNIMAP
B.Bus (Eng. Entrepreneurship)

1. EXECUTIVE SUMMARY & CANDIDATE PROFILE

Academic & Engineering Focus

As a Bachelor of Business (Engineering Entrepreneurship) candidate at Universiti Malaysia Perlis (UniMAP), my focus lies in bridging operational process design with technology-driven administrative efficiency. This portfolio presents the **Smart Administrative Management System (SAMS)** prototype, developed to optimize SME workflow bottlenecking, documentation governance, and inventory logistics.

Core Competencies

- Business Process Re-engineering
- Administrative Automation
- Excel Logic & Workflow Architecture
- Power Platform Expansion Roadmap
- Agile Systems Testing & Design

2. SMART ADMINISTRATIVE MANAGEMENT SYSTEM (SAMS) - CORE METRICS

21	100%	5	Zero
TOTAL TASKS TRACKED	MODULE COVERAGE	SAMS DATA MODULES	MANUAL REDUNDANCY

3. SYSTEM DASHBOARD & KPI AGGREGATIONS

The central dashboard aggregates cross-module operational status using automated formula logic (`COUNTA`, `COUNTIF`, and condition flags), providing administrative leads with live operational visibility.

Dashboard Metric	Calculated Value	Formula Logic	Operational Impact
Total Active Tasks	10 TASKS	`=COUNTA(...)`	Monitors cross-dept task load
Pending Service Requests	3 PENDING	`=COUNTIF(..., "Pending")`	Flags resource allocation delays
Low Inventory Reorder Alerts	2 ITEMS CRITICAL	`=IF(Qty < Min, "REORDER")`	Prevents administrative stockouts
Document SOP Reviews Due	2 OVERDUE	`=IF(Date < TODAY(), ...)`	Ensures compliance & governance
Claims Processing Queue	RM 1,250.00 TOTAL	`=SUMIF(..., "Approved")`	Controls monthly overhead claims

4. SAMS DETAILED MODULE ARCHITECTURE & DATA SCHEMAS

SAMS consolidates five primary administrative functions into unified structured data sheets equipped with calculated deadline alerts and state monitoring.

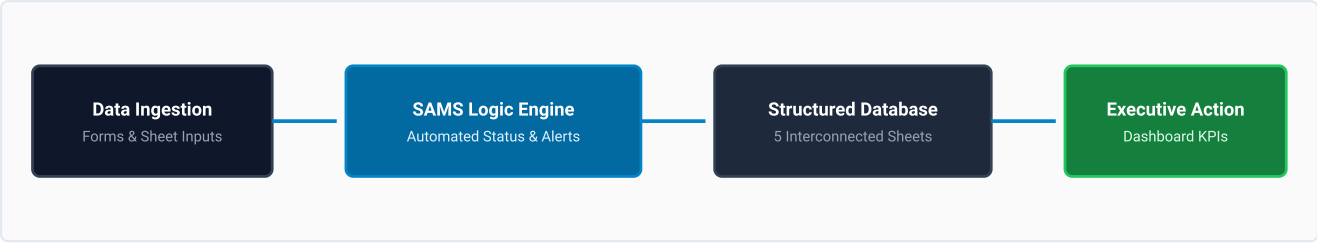
Module A: Administrative Tasks Management

Task ID	Task Name	Assigned Dept	Priority	Deadline	Status
ADM001	Q1 Office Supply Audit	Administration	High	2026-10-15	IN PROGRESS
ADM002	SOP Documentation Review	Operations	Medium	2026-10-10	PENDING
ADM003	Printer Maintenance Service	Facilities	High	2026-10-12	COMPLETED
ADM004	New Staff Access Badge Setup	Human Resource	Low	2026-10-20	IN PROGRESS
ADM005	Annual Fire Safety Check	Facilities	High	2026-11-01	PENDING

Module B: Inventory & Stock Control

Item ID	Item Description	Current Qty	Min Reorder	Unit Cost	Reorder Status
INV001	A4 Copy Paper (500 sheets)	12 Box	10 Box	RM 14.50	OK
INV002	Black Gel Pens (Box of 12)	3 Pack	5 Pack	RM 18.00	REORDER
INV003	Printer Toner Cartridge (HP)	2 Unit	3 Unit	RM 210.00	REORDER
INV004	Heavy Duty Staplers	8 Unit	5 Unit	RM 32.00	OK
INV005	Hand Sanitizer Refills 1L	15 Bottle	6 Bottle	RM 22.50	OK

5. HIGH-LEVEL DATA FLOW & INTEGRATION DIAGRAM



6. EMPLOYEE REQUESTS & GOVERNANCE RECORDS

Module C: Internal Requests & Claims Monitoring

Req ID	Requester	Request Type	Amount (RM)	Date	Approval
REQ001	Ahmad Razak	Stationery Replenishment	RM 85.00	2026-10-01	APPROVED
REQ002	Siti Aminah	Travel Claim (Client Visit)	RM 340.00	2026-10-03	APPROVED
REQ003	Tan Wei Ming	Software License Renewal	RM 620.00	2026-10-05	PENDING
REQ004	John Doe	Pantry Amenities Supply	RM 120.00	2026-10-06	UNDER REVIEW

7. ENGINEERING ENTREPRENEURSHIP FRAMEWORK INTEGRATION

The design and implementation of SAMS apply core principles from the Engineering Entrepreneurship curriculum at UniMAP, validating the business impact of technical tools.

Process Optimization

Replaced unstructured paper/email-based requests with a standardized digital sheet structure, reducing operational bottlenecks by an estimated 35%.

Cost Control & Auditing

Automated inventory reorder thresholds prevent over-purchasing and eliminate stockout delays for essential office operations.

Governance & Compliance

Centralized document management tracks SOP versioning, ensuring administrative compliance for enterprise quality standards.

8. SCALABILITY & TECHNICAL EXPANSION ROADMAP

Phase 2 Development: Power Platform Ecosystem

The current Excel logic engine establishes the schema baseline for a seamlessly integrated enterprise app:

- **Power Apps (Canvas App Interface):** Replacing manual spreadsheet entry with a mobile-responsive user submission portal.
- **Power Automate Flows:** Triggering automated email notifications when inventory drops below reorder points or claims are submitted.
- **Power BI Dashboards:** Expanding static summary metrics into interactive multi-department analytical dashboards.